CCF Database: A Machine-Learning-Annotated Corpus of 283,964 Canadian Climate Articles (1978–2026)

Supplementary Information

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This document collects the Supplementary Information accompanying the CCF Database manuscript. It contains: prior-literature and corpus composition tables (Supplementary Tables S1 and S2), the complete annotation framework with operational definitions (Supplementary Table S3), the complete per-category training metrics and dataset distribution (Supplementary Tables S4–S5), the Named Entity Recognition performance summary (Supplementary Table S6), the per-category validation metrics on the 1,000-sentence gold standard (Supplementary Table S7), the per-category distribution of annotations across the database (Supplementary Table S8), and four additional tables added during revision: per-category inter-coder reliability (Supplementary Table S9), reliability tier assignment (Supplementary Table S10), per-model training provenance (Supplementary Table S11), and the complete data dictionary of the enriched database (Supplementary Table S12).

Reproducibility note. Tables S1, S2, S3, and S6 are static prose / lookup tables inlined directly in this LaTeX source (their content does not depend on the live data). Tables S4 to S12 are reproducibly generated by `Scripts/Annotation/17_generate_tables.py` from the canonical CSVs in the `Database/Training_data/` directory and the live `CCF_Database` (Table S8 only); the generator reuses the category-normalisation API exposed by `Scripts/Annotation/15_normalize_categories.py`. Re-running scripts 15, 16, and 17 followed by a fresh `pdflatex` pass regenerates every numerical value in this document from the underlying data.

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Supplementary Table S1. Previous studies on climate change coverage in Canadian newspapers.

Study	Newspaper(s)	Scope	Language	Time Period
Good [1]	The Globe and Mail	National	English	2007
	The National Post	National	English	
	The Toronto Star	National	English	
	The Calgary Herald	Regional	English	
	The Edmonton Journal	Regional	English	
	The Chronicle Herald	Regional	English	
	The Montreal Gazette	Regional	English	
	The Charlottetown Guardian	Regional	English	
	Halifax Daily News	Regional	English	
	The Ottawa Citizen	Regional	English	
	The Star Phoenix	Regional	English	
	The Telegram	Regional	English	
	The Times Colonist	Regional	English	
	The Vancouver Sun	Regional	English	
	The Winnipeg Sun	Regional	English	
	Yukon News	Regional	English	
Rowe [2]	The Globe and Mail	National	English	2007–2008
	The Toronto Star	National	English	
	The Calgary Herald	Regional	English	
	The Toronto Sun	Regional	English	
	The Montreal Gazette	Regional	English	
	The Ottawa Citizen	Regional	English	
Young and Dugas [3]	The Globe and Mail	National	English	1988–2008
	The National Post	National	English	
Anne DiFrancesco and Young [4]	The Globe and Mail	National	English	2008
	The National Post	National	English	
Ahchong and Dodds [5]	The Globe and Mail	National	English	1988–2007
	The Toronto Star	National	English	
Stoddart and Tindall [6]	The Globe and Mail	National	English	1999–2010
	The National Post	National	English	

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Supplementary Table S1 – *Continued from previous page*

Study	Newspaper(s)	Scope	Language	Time Period
Ford and King [7]	The Globe and Mail	National	English	1993–2013
	The Toronto Star	National	English	
Stoddart and Smith [8]	The Globe and Mail	National	English	2006–2010
	The National Post	National	English	
Stoddart et al. [9]	The Globe and Mail	National	English	1997–2010
	The National Post	National	English	
Barkemeyer et al. [10]	The Globe and Mail	National	English	2008
	The National Post	National	English	
	The Toronto Star	National	English	
	The Vancouver Sun	Regional	English	
	The Toronto Sun	Regional	English	
King et al. [11]	The Globe and Mail	National	English	2005–2015
	The National Post	National	English	
	The Toronto Star	National	English	
	The Calgary Herald	Regional	English	
	The Chronicle Herald	Regional	English	
	The Vancouver Sun	Regional	English	
	The Whitehorse Daily Star	Regional	English	
	Journal de Montréal	Regional	French	
Pillod [12]	The Globe and Mail	National	English	2008–2020
Brandt [13]	The Toronto Star	National	English	2015–2018
	The Calgary Herald	Regional	English	
	The Vancouver Sun	Regional	English	
	The Chronicle Herald	Regional	English	
	The Winnipeg Free Press	Regional	English	
	Times and Transcript	Regional	English	
Hase et al. [14]	The Globe and Mail	National	English	2006–2018
	The Toronto Star	National	English	
Sachdeva and McCaffrey [15]	n/a	n/a	English	1986–2016
Stoddart and Yang [16]	The Globe and Mail	National	English	2015–2020

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Supplementary Table S1 – *Continued from previous page*

Study	**Newspaper(s)**	**Scope**	**Language**	**Time Period**
	The National Post	National	English	
	The Telegraph	Regional	English	
	The Telegram	Regional	English	
	The Chronicle Herald	Regional	English	
	The Guardian	Regional	English	
Chen et al. [17]	n/a	n/a	English	2018–2021

Note: This table summarizes the newspaper sources, geographic scope, language, and time periods covered by previous studies on climate change media coverage in Canada. Most studies focus exclusively on English-language national newspapers, particularly The Globe and Mail and The National Post.

Supplementary Table S2. Newspapers included in the CCF database.

Province/Territory	Newspaper	Language	Avg. Weekday Circ. (2015)	Articles
National	The Globe and Mail	English	226,171	31,127
	The National Post	English	79,157	20,829
	The Toronto Star	English	192,084	48,616
	CBC.ca[b]	English	–	2,941
	Radio-Canada.ca[b]	French	–	3,054
Alberta	The Edmonton Journal	English	59,244	18,763
	The Calgary Herald	English	58,865	20,349
British Columbia	The Vancouver Sun	English	78,901	19,068
	The Times Colonist	English	47,556	12,111
Manitoba	The Winnipeg Free Press	English	67,503	13,129
New Brunswick	L'Acadie Nouvelle[c]	French	18,102	5,143
Newfoundland & Labrador	The Telegram	English	15,380	6,208
Nova Scotia	The Chronicle Herald	English	71,304	11,018
Ontario	The Toronto Sun	English	85,775	3,333
	Le Droit[c]	French	28,616	4,728
Quebec	Le Devoir[c]	French	28,729	13,685
	Le Journal de Montréal	French	175,220	5,790
	La Presse	French	98,657	6,917
	La Presse+[a]	French	–	11,636
	The Montreal Gazette	English	53,344	9,799
Saskatchewan	The Star Phoenix	English	31,081	8,018
Yukon	The Whitehorse Daily Star[c]	English	945	7,702
Total				**283,964**

[a] In 2013, La Presse launched a free digital tablet edition called La Presse+ which offers slightly different content. The corpus considers articles published both on the website and in the digital tablet edition. Duplicates were discarded.
[b] News websites of the Canadian public broadcaster, added to the corpus by the CCF Observatory; their coverage begins in 2023. No print circulation applies.

[c] Coverage window bounded by the access period of the source aggregators: Le Droit until December 2023, The Whitehorse Daily Star until May 2024, Le Devoir and L'Acadie Nouvelle until March 2025. All other outlets are covered through July 2026 (corpus cut-off of release v2.0.0).

Supplementary Table S3. Complete CCF annotation framework.

#	Category	Code	Description
MAIN FRAMES			
Economic Frame			
1	**Economic Frame (Primary Category)**	`economic_frame`	If the sentence refers to climate change as an economic issue, including its negative or positive impacts on the economy, the financial costs or benefits of climate action, or the carbon footprint of the economic or industrial sector. This label captures any framing of climate change in economic terms, such as effects on jobs, markets, industries, or public and private finances.
2	Negative impacts of climate change on the economy	`eco_neg_impact`	If the sentence refers to the negative effects of climate change on the economy, such as damage to infrastructure, loss of agricultural productivity, disruptions to trade, increased costs for businesses, or economic decline in sectors vulnerable to climate shifts. This includes any economic losses directly resulting from changing climate conditions.
3	Positive impacts of climate change on the economy	`eco_pos_impact`	If the sentence refers to the positive effects of climate change on the economy, such as new agricultural opportunities, the opening of new trade routes, or the potential for economic growth in sectors adapting to climate shifts. This includes any economic gains directly resulting from changing climate conditions.
4	Economic disadvantages of climate action	`eco_cost`	If the sentence refers to the negative economic impacts caused by climate policies or climate action, such as costs associated with transitioning to green energy, investments in climate adaptation, debt incurred from financing climate action, or short-term job losses in fossil fuel industries. This category captures financial burdens and challenges linked to climate action or policy.
5	Economic benefits of climate action	`eco_benefit`	If the sentence refers to the positive economic impacts resulting from climate policies or climate action, such as job creation in renewable energy sectors, investments in green technologies, economic growth driven by sustainability initiatives, or cost savings from energy efficiency. This category captures financial gains and opportunities linked to climate action or policy.
6	Carbon footprint of the economic sector	`eco_footprint`	If the sentence refers to the environmental or carbon footprint of the economic or industrial sectors, including emissions and resource use related to manufacturing, transportation, energy production, mining, agriculture, and other commercial or industrial activities. This label covers the ecological impact generated by economic and industrial processes, not the effects of climate change on these sectors.
Health Frame			

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
7	**Health Frame (Primary Category)**	`health_frame`	If the sentence refers to climate change as a public health issue, including its negative or positive impacts on physical or mental health, the health co-benefits of climate action (e.g., cleaner air, active transportation), or the carbon footprint of the health sector. This label captures any framing of climate change in relation to health outcomes, healthcare systems, or health-related consequences and solutions.
8	Negative impacts of climate change on health	`health_neg_impact`	If the sentence refers to the negative effects of climate change on human health, such as increased risks of infectious diseases (e.g., malaria, dengue), respiratory and cardiovascular illnesses due to air pollution and heat exposure, heat-related illnesses (e.g., heatstroke), and mental health disorders (e.g., anxiety, depression) linked to environmental changes.
9	*Positive impacts of climate change on health**	`health_pos_impact`	If the sentence refers to the potential positive effects of climate change on human health, such as reductions in cold-related illnesses and deaths, or increased opportunities for outdoor activity due to milder winters.
10	Health co-benefits of climate action	`health_cobenefit`	If the sentence refers to health benefits resulting from climate policies or actions, such as improved air quality, better nutrition, reduced respiratory or cardiovascular diseases, fewer premature deaths, and other positive health outcomes linked to climate mitigation and adaptation efforts.
11	*Carbon footprint of the health sector**	`health_footprint`	If the sentence refers to the environmental or carbon footprint of the health sector, including emissions and resource use associated with healthcare facilities, medical equipment, pharmaceuticals production, patient and staff travel, and waste generated by medical activities. This label covers the ecological impact of delivering health services, not the impacts of climate change on health outcomes.
Security Frame			
12	**Security Frame (Primary Category)**	`security_frame`	If the sentence refers to the effects of climate change on security, such as violent conflict driven by resource scarcity, an influx of climate refugees, disruption of military operations, or the carbon footprint of the defense sector. This category includes broader security challenges arising from climate change, including geopolitical tensions or the strategic responses to climate impacts.
13	Presence of climate refugees	`security_refugees`	If the sentence refers to the displacement of populations or the presence of climate refugees as a result of a natural disaster, environmental degradation, or climate change impact. This category captures the movement of people fleeing climate-related disasters.

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
14	Conflict	`security_conflict`	If the sentence refers to the emergence or intensification of conflicts over natural resource exploitation due to climate change, such as disputes over water, land, or energy resources exacerbated by shifting climate patterns. This includes conflicts arising from scarcity or competition for increasingly stressed resources.
15	*Post-disaster military assistance**	`security_military`	If the sentence refers to the deployment of military forces as reinforcement after a natural disaster, such as wildfires, floods, or other extreme events. This includes military support in disaster response and recovery, like providing humanitarian aid or maintaining order during a crisis.
16	*Disruption of military operations**	`security_disruption`	If the sentence refers to the disruption of military operations due to a natural disaster affecting a military base or facilities. This includes cases where climate-induced events damage or disrupt military infrastructure, affecting defense readiness or response capabilities.
Justice Frame			
17	**Justice Frame (Primary Category)**	`justice_frame`	If the sentence frames climate change as a justice or moral issue, including references to who benefits or loses from climate action, differentiated responsibilities across countries or groups, unequal impacts of climate change, disparities in access to mitigation or adaptation measures, or concerns about intergenerational justice. This label captures ethical, equity-based, or fairness-oriented dimensions of climate discourse.
18	Winners and losers of climate action	`justice_winners`	If the sentence refers to how climate policies or actions create winners and losers, by benefiting certain groups (e.g., vulnerable populations, Indigenous communities, developing countries) while disadvantaging others (e.g., fossil fuel workers, carbon-intensive industries, or developed nations facing steeper transitions). This label focuses on distributional outcomes of climate policy.
19	Differentiated responsibility	`justice_responsibility`	If the sentence refers to the unequal responsibility for causing climate change, especially in reference to the principle of common but differentiated responsibilities. This includes statements emphasizing that developed countries, multinational corporations, or high-emitting individuals have historically contributed more to global warming and should therefore bear a greater share of the mitigation burden.
20	Unequal vulnerability to climate change	`justice_vulnerability`	If the sentence highlights how different populations are unequally affected by the impacts of climate change, due to geographic, social, or economic vulnerabilities. This includes references to marginalized groups such as women, seniors, racialized populations, people in developing countries, or those living in high-risk zones (e.g., coastal or arid regions). This label focuses on impacts, not responsibility or capacity.

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
21	Unequal access to climate action	`justice_access`	If the sentence refers to unequal capacity to act on climate change, emphasizing that countries, companies, or individuals do not have equal financial, technical, or institutional means to take climate action. For example, small businesses may struggle to decarbonize compared to multinational firms, and low-income households may not afford sustainable alternatives. This label reflects inequality in ability to act, not impacts or blame.
22	Intergenerational justice	`justice_intergen`	If the sentence refers to intergenerational justice, meaning that current decisions and actions should not compromise the rights or well-being of future generations. Do not apply this label to sentences that merely mention age differences or generational groups (e.g., "young people are protesting") unless the statement clearly relates to moral responsibility across generations or the long-term consequences of climate inaction.
Political Frame			
23	**Political Frame (Primary Category)**	`political_frame`	If the sentence frames climate change as a political issue, including references to policy adoption or implementation, political debates or controversies, political positioning by parties or leaders on climate issues, and public opinion data related to climate change. This label captures the role of governance, decision-making, and political dynamics in climate discourse.
24	Policy action	`pol_action`	If the sentence refers to the formal approval, adoption, or enactment of a policy, plan, or strategy by a government at the local, provincial, national, or international level. This includes the passing of new legislation, regulations, government programs, or official commitments related to climate or environmental issues (e.g., approval of a carbon pricing law, ratification of a renewable energy policy, adoption of a climate action framework).
25	Political debate	`pol_debate`	If the sentence refers to a political debate or discussion about climate change, particularly focusing on the differing views or disagreements on climate policies, the effectiveness of past actions, or the responsibility of different levels of government (e.g., federal vs. provincial). This includes disagreements about specific climate initiatives, such as proposed or enacted policies, laws, or funding allocations.
26	Political positioning	`pol_position`	If the sentence refers to the explicit stance or position of a politician or political party on climate change or climate-related policies. This includes statements where a political figure or party expresses their support or opposition to specific climate actions (e.g., carbon taxes, energy transitions, or climate mitigation strategies), or when they make electoral promises regarding climate policies.

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
27	Public opinion data	`pol_opinion`	If the sentence refers to data, surveys, polls, or studies measuring public opinion, attitudes, beliefs, or perceptions related to climate change, environmental policies, or climate action. This label captures quantitative or qualitative information on how the public views climate issues.
Scientific Frame			
28	**Scientific Frame (Primary Category)**	`scientific_frame`	If the sentence frames climate change as a scientific issue, including references to scientific research, discoveries, explanations, or assessments of climate change. This includes mentions of scientific consensus or controversy, as well as expressions of uncertainty or certainty in climate science. This label captures the role of scientific knowledge and expertise in understanding and communicating climate change.
29	Scientific debate	`sci_debate`	If the sentence refers to a past or ongoing debate or disagreement within the scientific community about aspects of climate science or climate-related technologies. This includes contested interpretations of data, disputes about methodologies, or ethical debates surrounding proposed scientific or technological solutions such as geoengineering. It does not include general questioning of climate science or statements that challenge the validity or sufficiency of climate evidence.
30	Popularisation or scientific discovery	`sci_discovery`	If the sentence explains or describes climate science mostly grounded in the natural sciences—such as chemistry, physics, or biology—or presents scientific or technological discoveries related to climate change. This includes explanations of mechanisms (e.g., the greenhouse effect, climate feedbacks), climate modeling, or innovations such as carbon capture. It does not apply to statements that merely describe environmental conditions or trends without explaining an underlying scientific process.
31	Questioning of climate science	`sci_skepticism`	If the sentence questions the validity, accuracy, or sufficiency of climate science. This includes claims that climate projections are flawed, evidence is lacking, or conclusions are exaggerated. It only applies when science is being challenged or delegitimized, not when scientists acknowledge limitations, complexity, or areas of ongoing research.
32	Defense of climate science	`sci_defense`	If the sentence defends the robustness or credibility of climate science in response to skepticism or denial. This includes assertions that reinforce the scientific consensus, affirm the validity of climate models, or rebut claims that climate science is incorrect or misleading. It only applies when climate science is defended, not when it is neutrally presented.
Environmental Frame			

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
33	**Environmental Frame (Primary Category)**	`environmental_frame`	If the sentence frames climate change as an environmental issue, including references to the degradation or disappearance of natural habitats, loss of biodiversity, and negative impacts on fauna and flora. This label captures ecological consequences of climate change and emphasizes the relationship between climate disruption and the natural environment.
34	Loss of natural environments	`env_habitat`	If the sentence refers to the long-term or irreversible disappearance or degradation of natural environments due to climate change. This includes situations where ecosystems or environmental features (e.g., glaciers, coral reefs) are predicted or expected to permanently disappear or significantly decline as a result of climate impacts, indicating that they will not recover. It does not apply to short-term, isolated, or recoverable events, such as one-time disasters (e.g., a single wildfire or a temporary drought) that might affect natural environments but do not necessarily imply permanent loss.
35	Loss of fauna and flora	`env_species`	If the sentence refers to the negative impacts of climate change on animal and plant species, including changes in habitat, shifts in migration or breeding patterns, increased mortality, loss of biodiversity, or threats to species survival. This label covers both temporary and ongoing effects on fauna and flora caused by climate change.
Cultural Frame			
36	**Cultural Frame (Primary Category)**	`cultural_frame`	If the sentence frames climate change as a cultural issue, including cultural depictions of climate change through art, literature, music, or film; challenges to hosting artistic or sports events due to climate impacts; the loss or disruption of Indigenous cultural practices; or the carbon footprint of the cultural and creative sectors. This label captures how climate change affects and is expressed through cultural life and identities.
37	Artistic representation	`cult_art`	If the sentence refers to cultural representations of climate change in various forms such as art, literature, music, film, documentaries, theater, or other creative expressions. This label captures how climate change is portrayed, interpreted, or communicated through cultural and artistic mediums.
38	Difficulty to host cultural or sports events	`cult_event_impact`	If the sentence refers to how climate change disrupts or challenges the organization of artistic, cultural, or sports events, such as cancelled festivals, concerts, or competitions due to extreme weather (e.g., storms, heatwaves, wildfires, or lack of snow for winter sports). This label captures climate-related obstacles to event planning or continuity.
39	Loss of indigenous practices	`cult_indigenous`	If the sentence refers to the erosion or disruption of traditional cultural practices among Indigenous communities due to climate change, such as changes in animal migration, plant cycles, or land access that affect food systems, ceremonies, or knowledge transmission. This label emphasizes climate-driven impacts on Indigenous cultural continuity.

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
40	Carbon footprint of the cultural and sports sectors	`cult_footprint`	If the sentence refers to the environmental or carbon footprint of the cultural, artistic, or sports sectors, including the emissions and resource use associated with activities such as touring, international travel, energy-intensive venues, unsustainable stage or set production, or event-related waste. This label covers the ecological impact of producing and hosting cultural or sports events, not the impacts of climate on those events.
PRIMARY CATEGORIES			
Actors/Messengers			
41	**Presence of Messengers (Primary Category)**	`messenger`	If the sentence includes mentions or quotes of persons or organizations by journalists, whether through direct quotations or indirect references. This label captures the presence of sources, experts, officials, activists, or other actors cited or named within the text. A single individual or organization may be associated with multiple types of expertise or roles (e.g., a doctor would fall both under Health expert and Natural scientist).
42	Health expert	`msg_health`	If the quoted or reported statement comes from a person or organization with medical or health-related recognized expertise. This includes individuals working in the fields of medicine, public health, healthcare administration, or biomedical research, whether in a professional, academic, or governmental capacity (e.g., professor of public health, doctor, nurse, family physician, public health officer, WHO representative, Red Cross medical coordinator, Doctors without Borders spokesperson, Canadian Medical Association representative, certified health consultants, minister of Health).
43	Economic expert	`msg_economic`	If the quoted or reported statement comes from a person or organization with economic or financial recognized expertise. This includes individuals working in the fields of economics, finance, business, or economic policy, whether in a professional, academic, private sector, or governmental capacity (e.g., economist, finance minister, central bank official, business executive, investment analyst, professor of economics, representative of a chamber of commerce, World Bank or IMF spokesperson, economic think tank expert, labour market analyst).
44	Security expert	`msg_security`	If the quoted or reported statement comes from a person or organization with recognized expertise in security or defence matters. This includes individuals with academic, professional, or operational experience in military, national security, intelligence, or strategic affairs (e.g., professor in defense or security studies, active or retired member of the armed forces, intelligence agency representative, conflict resolution specialist, national defence minister, security analyst, representative of a security-focused think tank or NGO).

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
45	Legal expert	`msg_legal`	If the quoted or reported statement comes from a person or organization with recognized expertise in law or legal affairs. This includes individuals working in legal professions, legal academics, the judiciary, or justice-related government roles (e.g., lawyer, prosecutor, law professor, minister of justice, attorney general, legal aid representative, bar association representative, legal consultants).
46	Cultural or Sport expert	`msg_cultural`	If the quoted or reported statement comes from a person with recognized expertise in sports, arts, or culture (e.g., Olympian, coach, professor of literature, museum director, filmmaker, writer, festival organizer, musician, artist, indigenous knowledge keeper).
47	Natural scientist	`msg_scientist`	If the quoted or reported statement comes from a person or organization with recognized expertise in the natural or hard sciences. This includes individuals working in fields such as climatology, environmental science, physics, chemistry, biology, earth sciences, engineering, or related disciplines, whether in academic, governmental, or research institutions (e.g., climate scientist, environmental researcher, oceanographer, physicist, ecologist, geologist, engineer, professor of atmospheric sciences, IPCC contributor, Environment Canada scientist, research institute spokesperson).
48	Social scientist	`msg_social`	If the quoted or reported statement comes from a person with expertise in social sciences, helping to understand how society perceives, responds to, and governs issues like climate change (e.g., sociologist, political scientist, media scholar, economist).
49	Activist	`msg_activist`	If the quoted or reported statement comes from a person or organization known primarily for advocacy or activism. This includes individuals or groups engaged in campaigning, protesting, or raising awareness on environmental, social, or climate-related issues, whether through grassroots organizing, NGOs, or international movements (e.g., climate activist, environmental campaigner, Indigenous land defender, Fridays for Future organizer, Greenpeace spokesperson, Extinction Rebellion member, citizen activist, youth climate leader, environmental justice advocate).
50	Public official	`msg_official`	If the quoted or reported statement comes from a politician, elected official, government representative, or public decision-maker at any level of government. This includes individuals holding political office or involved in policymaking, regulation, or governance, whether municipal, provincial, federal, or international (e.g., prime minister, environment minister, member of parliament, city councillor, premier, senator, government spokesperson, cabinet member, UN climate negotiator, provincial regulator, parliamentary committee chair).
Events			

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#	Category	Code	Description
51	**Presence of Events (Primary Category)**	`event`	If the sentence refers to the occurrence or mention of one or more events. This includes extreme meteorological events (e.g., storms, floods, heatwaves), political or governmental events (e.g., elections, policy announcements, trials, judiciary decisions), meetings or conferences, publications, cultural events, protests, or other significant happenings.
52	Extreme meteorological event	`evt_weather`	If the sentence refers to the occurrence of an extreme meteorological or climate-related event that causes or risks causing significant disruption, damage, or harm. This includes events such as heatwaves, floods, wildfires, hurricanes, droughts, storms, or other weather-related disasters, whether sudden or prolonged.
53	Meeting	`evt_meeting`	If the sentence refers to an official visit, conference, summit, forum, meeting, convention, or assembly that brings together people to discuss, negotiate, or present on specific topics at the local, national, or international levels (e.g., town hall meeting, premiers' conference, ministerial roundtable, state visit, UN conference, international forum on clean energy, G7 meeting, COP28, World Economic Forum).
54	Publication	`evt_publication`	If the sentence refers to the release or publication of a document such as a report, academic article, survey results, op-ed, policy brief, or white paper, often aimed at informing the public, guiding decision-making, or contributing to public discourse (e.g., IPCC report, Statistics Canada survey, university study on climate impacts, editorial on climate policy, NGO report on emissions, think tank publication).
55	Election	`evt_election`	If the sentence refers to the occurrence of an election at the local, provincial, national, or international level, including general elections, by-elections, leadership races, referenda, or electoral campaigns (e.g., federal election, municipal by-election, provincial leadership race, EU parliamentary elections).
56	New policy	`evt_policy`	If the sentence refers to the announcement or unveiling of a policy, plan, or strategy at the local, provincial, national, or international level, including new legislation, regulations, government programs, or official commitments, including those related to climate or environmental issues (e.g., carbon pricing plan, renewable energy strategy, climate action framework).
57	Judiciary decision	`evt_judiciary`	If the sentence refers to the occurrence of a trial, court ruling, legal proceeding, or judicial decision at any level of the judicial system, including constitutional challenges, environmental lawsuits, regulatory hearings, or Supreme Court decisions, including those related to climate or environmental issues (e.g., court ruling on carbon pricing, environmental class action lawsuit, constitutional challenge to climate legislation, tribunal decision on pipeline approval).

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
58	Cultural or Sports event	`evt_cultural`	If the sentence refers to the organization or hosting of a sports, artistic, or cultural event (e.g., Olympics games, national hockey tournament, local marathon, film screening, music concert, theatre performance, book launch, mural festival, photography exhibit).
59	Protest	`evt_protest`	If the sentence refers to the organization of a protest (e.g., climate strike, anti-pipeline protest, extinction rebellion action, anti-racism march, women's right rally, union-led protest for better wages).
Solutions			
60	**Presence of Solutions (Primary Category)**	`solution`	If the sentence refers to solutions addressing climate change, including mitigation efforts to reduce greenhouse gas emissions or adaptation strategies to manage climate impacts. This includes direct mentions of specific policies, technologies, behaviors, or initiatives, as well as indirect references to approaches aimed at combating or coping with climate change.
61	Mitigation strategy	`sol_mitigation`	If the sentence refers to mitigation solutions designed to reduce or prevent the emission of greenhouse gases and limit the extent of future climate change. These strategies aim to address the root causes of climate change by lowering carbon footprints, enhancing carbon sinks, or transitioning to low-carbon technologies (e.g., investing in renewable energy, improving energy efficiency, implementing carbon pricing, promoting public transit, reforestation, phasing out fossil fuels).
62	Adaptation strategy	`sol_adaptation`	If the sentence refers to adaptation solutions designed to reduce vulnerability to the negative effects of climate change by adjusting or preparing for those impacts. These strategies involve actions to manage the risks and enhance resilience, particularly by reducing exposure to climate-related hazards and minimizing harm to people, infrastructure, and ecosystems (e.g., providing cooling centres during heat waves, restoring wetlands to limit flooding, altering farming techniques to cope with changing weather patterns).
EMOTIONAL TONE			
63	Emotion: positive	`tone_positive`	If the tone is generally optimistic, reassuring, or enthusiastic, highlighting favorable aspects or expressing confidence in climate solutions or actions.
64	Emotion: negative	`tone_negative`	If the tone is generally alarming, critical, or pessimistic, especially if the sentence expresses concern, fear, or frustration about climate change or its impacts.
65	Emotion: neutral	`tone_neutral`	If the sentence is mainly informative or descriptive without an evaluative tone, presenting facts or balanced perspectives without strong emotional content.
GEOGRAPHIC FOCUS			

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Supplementary Table S3 – *Continued from previous page*

#	Category	Code	Description
66	Mention of Canada	`canada`	If the sentence refers to Canada in any context, domestic or international, including but not limited to climate change impacts on Canadian territory, as well as Canada's policies, actions, or roles.
URGENCY/ALARMISM			
67	Urgency to act	`urgency`	If the sentence conveys a strong sense of urgency or alarmism, emphasizing the immediate need for action or highlighting critical risks and dangers related to climate change, often with a tone that signals warning or crisis.

* Categories marked with an asterisk were included in the annotation protocol but had insufficient positive training examples. This table enumerates the 67 *candidate* categories of the framework; the deployed database carries 65 of them. Two candidates (`health_pos_impact`, `health_footprint`) had no positive examples in either language and were excluded entirely. Two deployed categories (`security_military`, `security_disruption`) could not be trained in English and are annotated on French sentences only (their English annotations are `NULL` by construction). The deployed model suite therefore counts $65 \times 2 - 2 = 128$ models. Because this table numbers the 67 candidates while `CCF_reliability_tiers` numbers the 65 deployed categories, the two numbering schemes differ from the Health frame onward: always match categories on the `code` column, never on the row number.

Supplementary Table S4. Complete model training performance metrics for all annotation categories.

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
MAIN FRAMES								
Economic Frame								
1	**Economic Frame (Primary Category)**	`economic_frame`	0.745	0.814	0.944	0.957	0.845	0.885
2	Negative impacts of climate change on the economy	`eco_neg_impact`	0.727	0.889	0.914	0.944	0.821	0.917
3	Positive impacts of climate change on the economy	`eco_pos_impact`	0.333	0.400	0.995	0.996	0.664	0.698
4	Economic disadvantages of climate action	`eco_cost`	0.615	0.500	0.848	0.913	0.732	0.707
5	Economic benefits of climate action	`eco_benefit`	0.500	0.516	0.952	0.981	0.726	0.749
6	Carbon footprint of the economic sector	`eco_footprint`	0.857	0.857	0.938	0.950	0.897	0.904
Health Frame								
7	**Health Frame (Primary Category)**	`health_frame`	0.800	0.667	0.989	0.994	0.894	0.830
8	Negative impacts of climate change on health	`health_neg_impact`	0.909	0.857	0.000	0.000	0.455	0.429
9	Health co-benefits of climate action	`health_cobenefit`	0.400	0.011	0.571	0.227	0.486	0.119
Security Frame								
10	**Security Frame (Primary Category)**	`security_frame`	0.870	0.800	0.996	0.997	0.933	0.898
11	Presence of climate refugees	`security_refugees`	1.000	1.000	1.000	1.000	1.000	1.000

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	**FR**	**EN**	**FR**	**EN**	**FR**
12	Conflict	`security_conflict`	1.000	1.000	1.000	1.000	1.000	1.000
13	Post-disaster military assistance	`security_military`	–	0.010	–	0.043	–	0.026
14	Disruption of military operations	`security_disruption`	–	1.000	–	1.000	–	1.000
Justice Frame								
15	**Justice Frame (Primary Category)**	`justice_frame`	0.719	0.717	0.975	0.981	0.847	0.849
16	Winners and losers of climate action	`justice_winners`	0.667	0.667	0.933	0.923	0.800	0.795
17	Differentiated responsibility	`justice_responsibility`	0.857	0.750	0.909	0.750	0.883	0.750
18	Unequal vulnerability to climate change	`justice_vulnerability`	0.571	0.600	0.992	0.995	0.782	0.798
19	Unequal access to climate action	`justice_access`	0.364	0.625	0.000	0.993	0.182	0.809
20	Intergenerational justice	`justice_intergen`	0.933	0.800	0.999	0.999	0.966	0.899
Political Frame								
21	**Political Frame (Primary Category)**	`political_frame`	0.808	0.774	0.897	0.888	0.853	0.831
22	Policy action	`pol_action`	0.621	0.667	0.985	0.966	0.803	0.816
23	Political debate	`pol_debate`	0.916	0.966	0.222	0.571	0.569	0.768
24	Political positioning	`pol_position`	0.750	0.800	0.946	0.989	0.848	0.895
25	Public opinion data	`pol_opinion`	0.909	1.000	0.999	1.000	0.954	1.000
Scientific Frame								
26	**Scientific Frame (Primary Category)**	`scientific_frame`	0.784	0.702	0.953	0.962	0.869	0.832
27	Scientific debate	`sci_debate`	0.737	0.500	0.848	0.875	0.793	0.688

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	FR	EN	FR	EN	FR
28	Popularisation or scientific discovery	sci_discovery	0.828	0.938	0.800	0.750	0.814	0.844
29	Questioning of climate science	sci_skepticism	0.727	0.333	0.927	0.995	0.827	0.664
30	Defense of climate science	sci_defense	0.571	0.000	0.933	0.919	0.752	0.459
Environmental Frame								
31	**Environmental Frame (Primary Category)**	environmental_frame	0.842	0.625	0.989	0.980	0.915	0.802
32	Loss of natural environments	env_habitat	1.000	0.857	1.000	0.800	1.000	0.829
33	Loss of fauna and flora	env_species	0.889	0.857	0.857	0.800	0.873	0.829
Cultural Frame								
34	**Cultural Frame (Primary Category)**	cultural_frame	0.773	0.833	0.986	0.993	0.879	0.913
35	Artistic representation	cult_art	1.000	1.000	1.000	1.000	1.000	1.000
36	Difficulty to host cultural or sports events	cult_event_impact	0.706	1.000	0.993	1.000	0.850	1.000
37	Loss of indigenous practices	cult_indigenous	1.000	0.026	1.000	0.899	1.000	0.462
38	Carbon footprint of the cultural and sports sectors	cult_footprint	1.000	0.005	1.000	0.143	1.000	0.074
PRIMARY CATEGORIES								
Actors/Messengers								
39	**Presence of Messengers (Primary Category)**	messenger	0.912	0.929	0.904	0.915	0.908	0.922
40	Health expert	msg_health	0.857	0.909	0.997	0.999	0.927	0.954
41	Economic expert	msg_economic	0.600	0.750	0.970	0.973	0.785	0.861
42	Security expert	msg_security	0.667	1.000	0.993	1.000	0.830	1.000

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	**FR**	**EN**	**FR**	**EN**	**FR**
43	Legal expert	`msg_legal`	0.667	0.769	0.999	0.996	0.833	0.883
44	Cultural or Sport expert	`msg_cultural`	1.000	0.556	1.000	0.990	1.000	0.773
45	Natural scientist	`msg_scientist`	0.789	0.833	0.977	0.971	0.883	0.902
46	Social scientist	`msg_social`	0.769	0.645	0.996	0.986	0.883	0.816
47	Activist	`msg_activist`	0.571	1.000	0.978	1.000	0.775	1.000
48	Public official	`msg_official`	0.703	0.923	0.897	0.964	0.800	0.943
Events								
49	**Presence of Events (Primary Category)**	`event`	0.794	0.819	0.935	0.932	0.865	0.876
50	Extreme meteorological event	`evt_weather`	1.000	0.857	1.000	0.968	1.000	0.912
51	Meeting	`evt_meeting`	0.824	0.957	0.936	0.982	0.880	0.969
52	Publication	`evt_publication`	0.873	1.000	0.990	1.000	0.931	1.000
53	Election	`evt_election`	1.000	0.800	1.000	0.998	1.000	0.899
54	New policy	`evt_policy`	0.769	0.727	0.943	0.954	0.856	0.841
55	Judiciary decision	`evt_judiciary`	1.000	1.000	1.000	1.000	1.000	1.000
56	Cultural or Sports event	`evt_cultural`	0.333	1.000	0.995	1.000	0.664	1.000
57	Protest	`evt_protest`	0.889	0.727	0.999	0.996	0.944	0.862
Solutions								
58	**Presence of Solutions (Primary Category)**	`solution`	0.737	0.878	0.914	0.944	0.825	0.911
59	Mitigation strategy	`sol_mitigation`	0.750	0.812	0.935	0.942	0.842	0.877
60	Adaptation strategy	`sol_adaptation`	0.696	0.800	0.991	0.989	0.843	0.894
EMOTIONAL TONE								
61	Emotion: positive	`tone_positive`	0.526	0.690	0.966	0.968	0.746	0.829
62	Emotion: negative	`tone_negative`	0.706	0.765	0.785	0.843	0.745	0.804
63	Emotion: neutral	`tone_neutral`	0.741	0.789	0.676	0.693	0.709	0.741
GEOGRAPHIC FOCUS								

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Supplementary Table S4 – Continued from previous page

#	Category	Code	F1 (Class 1)		F1 (Class 0)		Macro F1	
			EN	**FR**	**EN**	**FR**	**EN**	**FR**
64	Mention of Canada	`canada`	0.942	0.964	0.968	0.980	0.955	0.972
URGENCY TO ACT								
65	Urgency to act	`urgency`	0.591	0.649	0.975	0.984	0.783	0.816
AVERAGE PERFORMANCE METRICS								
		English Average	0.769	–	0.905	–	0.837	–
		French Average	–	0.737	–	0.894	–	0.816
		Overall Average	0.753		0.900		0.826	
TOTAL: 65 ANNOTATION CATEGORIES *(65 categories with at least one model; 0 categories entirely excluded*)*								

Note. F1 scores at the best epoch retained by the combined-criterion best-model selector ($0.7 \cdot F_1^{(1)} + 0.3\cdot$ macro-F_1). EN columns report metrics for the English BERT-base-cased classifiers; FR columns report metrics for the French CamemBERT-base classifiers. A "–" in one language column means the category could not be trained in that language for lack of positive examples: this concerns exactly two English models (`security_military`, `security_disruption`), which is why the deployed suite counts $65 \times 2 - 2 = 128$ models; the two candidate categories excluded in both languages do not appear in this table (see the note to Supplementary Table S3). The dataset distribution behind each row is reported in Supplementary Table S5.

Supplementary Table S5. Training and validation dataset distribution across all annotation categories.

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
MAIN FRAMES										
Economic Frame										
1	**Economic Frame (Primary Category)**	`economic_frame`	218	1067	24	118	252	1164	28	129
2	Negative impacts of climate change on the economy	`eco_neg_impact`	61	158	6	17	75	178	8	19
3	Positive impacts of climate change on the economy	`eco_pos_impact`	14	205	1	22	7	245	1	27
4	Economic disadvantages of climate action	`eco_cost`	63	156	6	17	42	211	4	23
5	Economic benefits of climate action	`eco_benefit`	36	183	3	20	48	205	5	22
6	Carbon footprint of the economic sector	`eco_footprint`	67	152	7	16	77	176	8	19
Health Frame										
7	**Health Frame (Primary Category)**	`health_frame`	57	1228	6	136	37	1379	4	153
8	Negative impacts of climate change on health	`health_neg_impact`	47	10	5	1	32	5	3	1
9	Health co-benefits of climate action	`health_cobenefit`	8	49	1	5	4	33	1	3
Security Frame										
10	**Security Frame (Primary Category)**	`security_frame`	19	1266	2	140	19	1397	2	155

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English Training Pos	English Training Neg	English Validation Pos	English Validation Neg	French Training Pos	French Training Neg	French Validation Pos	French Validation Neg
11	Presence of climate refugees	`security_refugees`	10	9	1	1	7	12	1	1
12	Conflict	`security_conflict`	6	13	1	1	7	12	1	1
13	Post-disaster military assistance	`security_military`	0	18	1	2	4	15	1	1
14	Disruption of military operations	`security_disruption`	0	18	1	2	1	18	1	1
Justice Frame										
15	**Justice Frame (Primary Category)**	`justice_frame`	90	1196	9	132	81	1335	9	148
16	Winners and losers of climate action	`justice_winners`	20	70	2	7	20	62	2	6
17	Differentiated responsibility	`justice_responsibility`	38	52	4	5	41	41	4	4
18	Unequal vulnerability to climate change	`justice_vulnerability`	8	81	1	9	9	72	1	8
19	Unequal access to climate action	`justice_access`	21	69	2	7	20	62	2	6
20	Intergenerational justice	`justice_intergen`	14	76	1	8	6	75	1	8
Political Frame										
21	**Political Frame (Primary Category)**	`political_frame`	416	869	46	96	440	977	48	108
22	Policy action	`pol_action`	62	355	6	39	58	382	6	42
23	Political debate	`pol_debate`	344	72	38	8	396	45	43	4
24	Political positioning	`pol_position`	61	356	6	39	35	405	3	45
25	Public opinion data	`pol_opinion`	21	396	2	43	20	420	2	46
Scientific Frame										

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
26	**Scientific Frame (Primary Category)**	`scientific_frame`	243	1042	27	115	185	1232	20	136
27	Scientific debate	`sci_debate`	98	146	10	16	46	139	5	15
28	Popularisation or scientific discovery	`sci_discovery`	126	117	14	13	144	41	16	4
29	Questioning of climate science	`sci_skepticism`	50	194	5	21	17	169	1	18
30	Defense of climate science	`sci_defense`	37	207	4	22	32	153	3	17
Environmental Frame										
31	**Environmental Frame (Primary Category)**	`environmental_frame`	84	1201	9	133	63	1354	6	150
32	Loss of natural environments	`env_habitat`	50	35	5	3	35	28	3	3
33	Loss of fauna and flora	`env_species`	42	43	4	4	32	31	3	3
Cultural Frame										
34	**Cultural Frame (Primary Category)**	`cultural_frame`	44	1242	4	137	45	1371	5	152
35	Artistic representation	`cult_art`	20	24	2	2	18	27	2	3
36	Difficulty to host cultural or sports events	`cult_event_impact`	11	33	1	3	21	25	2	2
37	Loss of indigenous practices	`cult_indigenous`	10	34	1	3	4	41	1	4
38	Carbon footprint of the cultural and sports sectors	`cult_footprint`	1	42	1	4	2	43	1	4
PRIMARY CATEGORIES										
Actors/Messengers										

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
39	**Presence of Messengers (Primary Category)**	`messenger`	657	629	72	69	736	681	81	75
40	Health expert	`msg_health`	10	647	1	71	8	728	1	80
41	Economic expert	`msg_economic`	55	602	6	66	67	669	7	74
42	Security expert	`msg_security`	3	653	1	72	10	726	1	80
43	Legal expert	`msg_legal`	1	655	1	72	4	731	1	81
44	Cultural or Sport expert	`msg_cultural`	13	644	1	71	9	727	1	80
45	Natural scientist	`msg_scientist`	107	550	11	61	117	620	12	68
46	Social scientist	`msg_social`	9	648	1	71	26	711	2	78
47	Activist	`msg_activist`	53	604	5	67	55	681	6	75
48	Public official	`msg_official`	171	486	18	54	223	513	24	57
Events										
49	**Presence of Events (Primary Category)**	`event`	298	987	33	109	355	1062	39	117
50	Extreme meteorological event	`evt_weather`	72	226	8	25	59	297	6	32
51	Meeting	`evt_meeting`	75	224	8	24	100	255	11	28
52	Publication	`evt_publication`	74	225	8	24	114	242	12	26
53	Election	`evt_election`	16	283	1	31	18	337	2	37
54	New policy	`evt_policy`	55	243	6	27	60	296	6	32
55	Judiciary decision	`evt_judiciary`	10	288	1	32	10	345	1	38
56	Cultural or Sports event	`evt_cultural`	2	296	1	32	11	344	1	38
57	Protest	`evt_protest`	11	288	1	31	8	347	1	38
Solutions										
58	**Presence of Solutions (Primary Category)**	`solution`	314	972	34	107	415	1001	46	111
59	Mitigation strategy	`sol_mitigation`	279	35	31	3	360	55	40	6

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Supplementary Table S5 – Continued from previous page

#	Category	Code	English				French			
			Training		Validation		Training		Validation	
			Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
60	Adaptation strategy	`sol_adaptation`	46	268	5	29	64	351	7	39
EMOTIONAL TONE										
61	Emotion: positive	`tone_positive`	103	1182	11	131	151	1266	16	140
62	Emotion: negative	`tone_negative`	491	794	54	88	537	880	59	97
63	Emotion: neutral	`tone_neutral`	686	599	76	66	726	691	80	76
GEOGRAPHIC FOCUS										
64	Mention of Canada	`canada`	445	840	49	93	493	924	54	102
URGENCY TO ACT										
65	Urgency to act	`urgency`	66	1219	7	135	63	1354	6	150

Note. Positive (**Pos**) and negative (**Neg**) sentence counts in the training and validation splits, by language. Categories with zero counts in every split could not be trained due to insufficient positive examples; the resulting per-category training metrics are reported in Supplementary Table S4.

Supplementary Table S6. Named Entity Recognition performance.

Table S11: Named Entity Recognition (`ner_entities` variable) model performance metrics

Language	**Model**	**PER F1**	**ORG F1**	**LOC F1**
English	BERT-base-NER	0.961	0.811	0.925
French	spaCy fr_core_news_lg	0.880	–	–
	CamemBERT-NER	–	0.824	0.929

Note: The hybrid approach for French combines spaCy for person entities (PER) with CamemBERT-NER for organization (ORG) and location (LOC) entities based on empirical evaluation on the dataset. Performance metrics are from the original model documentation.

Supplementary Table S7. Detailed validation performance metrics on the 1,000-sentence gold standard.

#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
MAIN FRAMES														
Economic Frame														
1	**Economic Frame (Primary Category)**	economic_frame	0.847	0.814	0.830	0.892	0.864	0.878	0.890	0.862	0.875	120	129	249
2	Negative impacts of climate change on the economy	eco_neg_impact	0.820	0.741	0.783	0.848	0.810	0.828	0.856	0.833	0.842	26	18	44
3	Positive impacts of climate change on the economy	eco_pos_impact	0.806	0.508	0.684	0.905	0.845	0.873	0.913	0.908	0.900	12	1	13
4	Economic disadvantages of climate action	eco_cost	0.741	0.702	0.723	0.800	0.802	0.801	0.813	0.806	0.810	22	23	45
5	Economic benefits of climate action	eco_benefit	0.885	0.817	0.848	0.924	0.871	0.896	0.928	0.872	0.899	19	26	45
6	Carbon footprint of the economic sector	eco_footprint	0.785	0.788	0.787	0.838	0.845	0.842	0.831	0.843	0.837	30	29	59
Health Frame														
7	**Health Frame (Primary Category)**	health_frame	0.800	0.817	0.808	0.947	0.961	0.954	0.947	0.964	0.956	35	23	58
8	Negative impacts of climate change on health	health_neg_impact	0.386	0.340	0.364	0.629	0.514	0.571	0.485	0.349	0.416	22	18	40
9	Health co-benefits of climate action	health_cobenefit	0.252	0.103	0.186	0.257	0.114	0.186	0.301	0.023	0.177	4	4	8
Security Frame														
10	**Security Frame (Primary Category)**	security_frame	0.825	0.730	0.782	0.951	0.939	0.945	0.955	0.950	0.952	30	19	49
11	Presence of climate refugees	security_refugees	0.476	0.417	0.481	0.545	0.429	0.488	0.468	0.476	0.459	21	6	27
12	Conflict	security_conflict	0.358	0.286	0.324	0.364	0.286	0.326	0.398	0.286	0.346	7	6	13
13	Post-disaster military assistance	security_military	0.000	0.045	0.045	0.000	0.048	0.048	0.000	0.004	0.004	0	2	2
14	Disruption of military operations	security_disruption	0.000	1.000	1.000	0.000	1.000	1.000	0.000	1.000	1.000	0	2	2
Justice Frame														
15	**Justice Frame (Primary Category)**	justice_frame	0.832	0.880	0.857	0.917	0.937	0.927	0.921	0.939	0.930	62	74	136

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Supplementary Table S7 – Continued from previous page

#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
16	Winners and losers of climate action	`justice_winners`	0.654	0.737	0.708	0.765	0.762	0.764	0.802	0.775	0.786	10	22	32
17	Differentiated responsibility	`justice_responsibility`	0.900	0.823	0.860	0.926	0.833	0.879	0.927	0.838	0.883	19	27	46
18	Unequal vulnerability to climate change	`justice_vulnerability`	0.786	0.894	0.845	0.852	0.917	0.885	0.858	0.918	0.888	16	21	37
19	Unequal access to climate action	`justice_access`	0.475	0.796	0.651	0.481	0.833	0.661	0.456	0.835	0.674	27	23	50
20	Intergenerational justice	`justice_intergen`	0.924	0.952	0.942	0.963	0.964	0.964	0.965	0.965	0.965	10	19	29
Political Frame														
21	**Political Frame (Primary Category)**	`political_frame`	0.812	0.842	0.829	0.827	0.844	0.836	0.829	0.845	0.837	167	217	384
22	Policy action	`pol_action`	0.802	0.722	0.756	0.914	0.864	0.886	0.909	0.854	0.879	26	38	64
23	Political debate	`pol_debate`	0.469	0.650	0.564	0.694	0.820	0.763	0.595	0.781	0.699	127	175	302
24	Political positioning	`pol_position`	0.752	0.808	0.782	0.801	0.912	0.862	0.821	0.920	0.877	35	24	59
25	Public opinion data	`pol_opinion`	0.877	0.910	0.894	0.962	0.974	0.969	0.966	0.976	0.971	12	15	27
Scientific Frame														
26	**Scientific Frame (Primary Category)**	`scientific_frame`	0.891	0.890	0.890	0.951	0.949	0.950	0.948	0.947	0.947	74	75	149
27	Scientific debate	`sci_debate`	0.962	0.867	0.911	0.962	0.869	0.912	0.962	0.869	0.912	26	26	52
28	Popularisation or scientific discovery	`sci_discovery`	0.884	0.797	0.838	0.885	0.803	0.841	0.886	0.803	0.842	31	36	67
29	Questioning of climate science	`sci_skepticism`	0.783	0.768	0.775	0.846	0.836	0.841	0.860	0.857	0.858	9	9	18
30	Defense of climate science	`sci_defense`	0.769	0.460	0.682	0.846	0.852	0.850	0.862	0.785	0.843	8	9	17
Environmental Frame														
31	**Environmental Frame (Primary Category)**	`environmental_frame`	0.866	0.875	0.871	0.967	0.967	0.967	0.969	0.968	0.969	28	32	60
32	Loss of natural environments	`env_habitat`	0.804	0.573	0.721	0.806	0.707	0.753	0.802	0.637	0.730	16	26	42
33	Loss of fauna and flora	`env_species`	0.714	0.692	0.703	0.722	0.707	0.714	0.717	0.694	0.705	19	21	40
Cultural Frame														
34	**Cultural Frame (Primary Category)**	`cultural_frame`	0.802	0.581	0.708	0.923	0.880	0.901	0.931	0.920	0.921	41	11	52
35	Artistic representation	`cult_art`	0.834	0.469	0.659	0.851	0.559	0.704	0.858	0.649	0.739	18	6	24
36	Difficulty to host cultural or sports events	`cult_event_impact`	0.631	0.354	0.484	0.731	0.456	0.593	0.777	0.595	0.686	8	2	10
37	Loss of indigenous practices	`cult_indigenous`	0.824	0.415	0.595	0.910	0.618	0.763	0.922	0.749	0.825	7	1	8

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Supplementary Table S7 – Continued from previous page

#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
38	Carbon footprint of the cultural and sports sectors	`cult_footprint`	0.702	0.014	0.351	0.925	0.015	0.467	0.945	0.000	0.613	2	1	3
PRIMARY CATEGORIES														
Actors/Messengers														
39	**Presence of Messengers (Primary Category)**	`messenger`	0.955	0.967	0.961	0.957	0.969	0.963	0.957	0.969	0.963	301	312	613
40	Health expert	`msg_health`	0.862	0.951	0.907	0.964	0.987	0.975	0.966	0.987	0.976	19	21	40
41	Economic expert	`msg_economic`	0.847	0.895	0.874	0.954	0.961	0.957	0.949	0.963	0.957	30	28	58
42	Security expert	`msg_security`	0.728	0.813	0.787	0.970	0.958	0.964	0.973	0.962	0.968	7	14	21
43	Legal expert	`msg_legal`	0.861	0.927	0.914	0.990	0.980	0.985	0.989	0.982	0.986	7	19	26
44	Cultural or Sport expert	`msg_cultural`	0.853	0.825	0.843	0.954	0.967	0.961	0.954	0.969	0.961	26	13	39
45	Natural scientist	`msg_scientist`	0.920	0.896	0.907	0.960	0.941	0.951	0.960	0.944	0.952	44	46	90
46	Social scientist	`msg_social`	0.938	0.948	0.944	0.987	0.984	0.985	0.987	0.984	0.986	17	24	41
47	Activist	`msg_activist`	0.934	0.903	0.918	0.977	0.964	0.970	0.976	0.963	0.970	31	33	64
48	Public official	`msg_official`	0.927	0.930	0.929	0.934	0.935	0.934	0.934	0.935	0.934	104	113	217
Events														
49	**Presence of Events (Primary Category)**	`event`	0.881	0.912	0.897	0.886	0.913	0.900	0.885	0.913	0.900	201	222	423
50	Extreme meteorological event	`evt_weather`	0.887	0.891	0.889	0.935	0.955	0.946	0.938	0.958	0.949	29	21	50
51	Meeting	`evt_meeting`	0.855	0.891	0.877	0.908	0.914	0.911	0.914	0.918	0.916	30	50	80
52	Publication	`evt_publication`	0.950	0.962	0.958	0.973	0.968	0.970	0.973	0.969	0.971	30	62	92
53	Election	`evt_election`	0.911	0.895	0.903	0.962	0.964	0.963	0.960	0.963	0.962	25	22	47
54	New policy	`evt_policy`	0.821	0.837	0.829	0.886	0.914	0.901	0.895	0.915	0.906	29	33	62
55	Judiciary decision	`evt_judiciary`	0.628	0.939	0.808	0.903	0.982	0.946	0.927	0.981	0.951	6	19	25
56	Cultural or Sports event	`evt_cultural`	0.906	0.857	0.881	0.973	0.964	0.968	0.971	0.967	0.968	17	12	29
57	Protest	`evt_protest`	0.982	0.909	0.943	0.995	0.973	0.983	0.995	0.973	0.983	16	18	34
Solutions														
58	**Presence of Solutions (Primary Category)**	`solution`	0.891	0.909	0.900	0.921	0.931	0.926	0.922	0.930	0.926	114	133	247
59	Mitigation strategy	`sol_mitigation`	0.872	0.801	0.839	0.884	0.839	0.861	0.882	0.850	0.864	76	97	173
60	Adaptation strategy	`sol_adaptation`	0.889	0.903	0.895	0.934	0.952	0.943	0.934	0.954	0.944	22	16	38
EMOTIONAL TONE														
61	Emotion: positive	`tone_positive`	0.712	0.692	0.702	0.902	0.892	0.897	0.910	0.904	0.907	38	37	75
62	Emotion: negative	`tone_negative`	0.793	0.804	0.799	0.807	0.813	0.810	0.814	0.819	0.816	151	165	316
63	Emotion: neutral	`tone_neutral`	0.756	0.798	0.777	0.764	0.809	0.787	0.765	0.807	0.787	295	306	601

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Supplementary Table S7 – Continued from previous page

#	Category	Code	F1 Macro			F1 Micro			F1 Weighted			Support		
			EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
GEOGRAPHIC FOCUS														
64	Mention of Canada	`canada`	0.986	0.978	0.982	0.986	0.978	0.982	0.986	0.978	0.982	235	255	490
URGENCY TO ACT														
65	Urgency to act	`urgency`	0.874	0.835	0.853	0.976	0.965	0.970	0.977	0.967	0.972	23	25	48
OVERALL PERFORMANCE														
		All Categories	**0.869**	**0.864**	**0.866**	**0.909**	**0.902**	**0.905**	**0.911**	**0.905**	**0.908**	**3069**	**3332**	**6401**

Note. F1 macro, F1 micro, F1 weighted, and positive-class support computed on the 1,000-sentence gold standard. EN, FR, and ALL columns report metrics on the English subset, the French subset, and their union. **OVERALL PERFORMANCE** in the last row reports the *pooled* aggregates—all per-category predictions are pooled into a single confusion matrix before the scores are computed—and is therefore not the mean of the per-category rows. The unweighted mean of the 65 per-category macro F1 values (ALL column) is 0.773; this is the quantity compared like-for-like with the training-phase mean of 0.826 in the main text (Technical Validation). Support columns report the number of expert-confirmed positive examples; 23 of the 65 categories fall below the sampling target of 40 (see the main text on how to weigh low-support estimates). Inter-coder agreement statistics on the same 1,000 sentences are reported in Supplementary Tables S9 and S10.

Supplementary Table S8. Database-wide distribution of annotation categories across the 9.9 million sentences of CCF_processed_data.

#	Category	Code	Count EN	Count FR	Count ALL	Proportion (%) EN	Proportion (%) FR	Proportion (%) ALL
MAIN FRAMES								
Economic Frame								
1	**Economic Frame (Primary Category)**	`economic_frame`	1,312,306	206,394	1,518,700	15.88	12.56	15.33
2	Negative impacts of climate change on the economy	`eco_neg_impact`	208,232	35,469	243,701	2.52	2.16	2.46
3	Positive impacts of climate change on the economy	`eco_pos_impact`	1,216	2,198	3,414	0.01	0.13	0.03
4	Economic disadvantages of climate action	`eco_cost`	549,771	45,273	595,044	6.65	2.75	6.01
5	Economic benefits of climate action	`eco_benefit`	79,448	32,961	112,409	0.96	2.01	1.13
6	Carbon footprint of the economic sector	`eco_footprint`	308,856	67,727	376,583	3.74	4.12	3.80
Health Frame								
7	**Health Frame (Primary Category)**	`health_frame`	116,449	24,852	141,301	1.41	1.51	1.43
8	Negative impacts of climate change on health	`health_neg_impact`	116,449	24,852	141,301	1.41	1.51	1.43
9	Health co-benefits of climate action	`health_cobenefit`	68,998	23,026	92,024	0.83	1.40	0.93
Security Frame								
10	**Security Frame (Primary Category)**	`security_frame`	34,882	11,402	46,284	0.42	0.69	0.47

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
11	Presence of climate refugees	`security_refugees`	22,723	4,556	27,279	0.27	0.28	0.28
12	Conflict	`security_conflict`	21,758	9,969	31,727	0.26	0.61	0.32
13	Post-disaster military assistance	`security_military`	0	11,400	11,400	0.00	0.69	0.12
14	Disruption of military operations	`security_disruption`	0	4	4	0.00	0.00	0.00
Justice Frame								
15	**Justice Frame (Primary Category)**	`justice_frame`	290,014	22,364	312,378	3.51	1.36	3.15
16	Winners and losers of climate action	`justice_winners`	47,725	9,625	57,350	0.58	0.59	0.58
17	Differentiated responsibility	`justice_responsibility`	61,525	9,362	70,887	0.74	0.57	0.72
18	Unequal vulnerability to climate change	`justice_vulnerability`	13,845	2,423	16,268	0.17	0.15	0.16
19	Unequal access to climate action	`justice_access`	225,960	5,067	231,027	2.73	0.31	2.33
20	Intergenerational justice	`justice_intergen`	28,753	1,295	30,048	0.35	0.08	0.30
Political Frame								
21	**Political Frame (Primary Category)**	`political_frame`	2,548,832	447,833	2,996,665	30.84	27.25	30.24
22	Policy action	`pol_action`	219,650	21,051	240,701	2.66	1.28	2.43
23	Political debate	`pol_debate`	2,494,072	438,459	2,932,531	30.17	26.68	29.60
24	Political positioning	`pol_position`	901,142	52,613	953,755	10.90	3.20	9.63
25	Public opinion data	`pol_opinion`	48,852	8,857	57,709	0.59	0.54	0.58
Scientific Frame								

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count EN	Count FR	Count ALL	Proportion (%) EN	Proportion (%) FR	Proportion (%) ALL
26	**Scientific Frame (Primary Category)**	`scientific_frame`	545,323	86,067	631,390	6.60	5.24	6.37
27	Scientific debate	`sci_debate`	221,198	15,847	237,045	2.68	0.96	2.39
28	Popularisation or scientific discovery	`sci_discovery`	366,775	72,724	439,499	4.44	4.43	4.44
29	Questioning of climate science	`sci_skepticism`	67,259	2,250	69,509	0.81	0.14	0.70
30	Defense of climate science	`sci_defense`	50,380	0	50,380	0.61	0.00	0.51
Environmental Frame								
31	**Environmental Frame (Primary Category)**	`environmental_frame`	218,374	88,446	306,820	2.64	5.38	3.10
32	Loss of natural environments	`env_habitat`	113,064	74,042	187,106	1.37	4.51	1.89
33	Loss of fauna and flora	`env_species`	126,270	61,208	187,478	1.53	3.72	1.89
Cultural Frame								
34	**Cultural Frame (Primary Category)**	`cultural_frame`	306,791	96,179	402,970	3.71	5.85	4.07
35	Artistic representation	`cult_art`	212,176	49,753	261,929	2.57	3.03	2.64
36	Difficulty to host cultural or sports events	`cult_event_impact`	25,402	42,776	68,178	0.31	2.60	0.69
37	Loss of indigenous practices	`cult_indigenous`	7,890	21,917	29,807	0.10	1.33	0.30
38	Carbon footprint of the cultural and sports sectors	`cult_footprint`	13	96,068	96,081	0.00	5.85	0.97
PRIMARY CATEGORIES								
Actors/Messengers								
39	**Presence of Messengers (Primary Category)**	`messenger`	4,131,733	744,674	4,876,407	49.99	45.31	49.21

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count			Proportion (%)		
			EN	**FR**	**ALL**	**EN**	**FR**	**ALL**
40	Health expert	`msg_health`	51,198	9,371	60,569	0.62	0.57	0.61
41	Economic expert	`msg_economic`	229,368	84,782	314,150	2.78	5.16	3.17
42	Security expert	`msg_security`	19	11,079	11,098	0.00	0.67	0.11
43	Legal expert	`msg_legal`	7	5,968	5,975	0.00	0.36	0.06
44	Cultural or Sport expert	`msg_cultural`	130,523	20,405	150,928	1.58	1.24	1.52
45	Natural scientist	`msg_scientist`	299,413	84,448	383,861	3.62	5.14	3.87
46	Social scientist	`msg_social`	34,576	22,512	57,088	0.42	1.37	0.58
47	Activist	`msg_activist`	101,751	27,501	129,252	1.23	1.67	1.30
48	Public official	`msg_official`	1,372,377	253,098	1,625,475	16.60	15.40	16.40
Events								
49	**Presence of Events (Primary Category)**	`event`	1,531,574	312,091	1,843,665	18.53	18.99	18.61
50	Extreme meteorological event	`evt_weather`	230,498	49,519	280,017	2.79	3.01	2.83
51	Meeting	`evt_meeting`	418,662	100,715	519,377	5.07	6.13	5.24
52	Publication	`evt_publication`	248,794	81,486	330,280	3.01	4.96	3.33
53	Election	`evt_election`	67,362	30,735	98,097	0.81	1.87	0.99
54	New policy	`evt_policy`	409,064	47,286	456,350	4.95	2.88	4.61
55	Judiciary decision	`evt_judiciary`	38,999	7,415	46,414	0.47	0.45	0.47
56	Cultural or Sports event	`evt_cultural`	1,026	25,484	26,510	0.01	1.55	0.27
57	Protest	`evt_protest`	8,078	5,881	13,959	0.10	0.36	0.14
Solutions								
58	**Presence of Solutions (Primary Category)**	`solution`	1,938,846	347,250	2,286,096	23.46	21.13	23.07
59	Mitigation strategy	`sol_mitigation`	1,331,925	222,359	1,554,284	16.11	13.53	15.69
60	Adaptation strategy	`sol_adaptation`	54,500	22,611	77,111	0.66	1.38	0.78

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Supplementary Table S8 – Continued from previous page

#	Category	Code	Count			Proportion (%)		
			EN	FR	ALL	EN	FR	ALL
EMOTIONAL TONE								
61	Emotion: positive	`tone_positive`	678,078	209,435	887,513	8.20	12.74	8.96
62	Emotion: negative	`tone_negative`	3,161,855	579,082	3,740,937	38.25	35.24	37.75
63	Emotion: neutral	`tone_neutral`	5,218,406	1,185,185	6,403,591	63.14	72.12	64.63
GEOGRAPHIC FOCUS								
64	Mention of Canada	`canada`	3,644,510	656,257	4,300,767	44.09	39.93	43.40
URGENCY TO ACT								
65	Urgency to act	`urgency`	170,813	39,472	210,285	2.07	2.40	2.12
Total Sentences			**8,265,398**	**1,643,378**	**9,908,776**	**100.00**	**100.00**	**100.00**

Note. Counts and proportions are computed live from `CCF_Database.CCF_processed_data` at table-build time; the "ALL" columns aggregate English and French. Proportions are expressed in percent of the total number of sentences in each language. The total sentence counts at the bottom of the table match the data-dictionary entry for `CCF_processed_data` in Supplementary Table S12.

Supplementary Table S9. Per-category inter-coder reliability metrics on the 1,000-sentence gold standard, with the blind phase (sentences 601–1000) reported separately.

#	Category	Code	Cohen's κ (full)	Blind phase (sentences 601–1000) κ	Krippendorff α	Gwet AC1	% agree	n
MAIN FRAMES								
Economic Frame								
1	**Economic Frame (Primary Category)**	`economic_frame`	0.605	0.625	0.625	0.735	0.868	400
2	Negative impacts of climate change on the economy	`eco_neg_impact`	0.694	0.542	0.542	0.915	0.958	400
3	Positive impacts of climate change on the economy	`eco_pos_impact`	0.928	0.831	0.831	0.990	0.995	400
4	Economic disadvantages of climate action	`eco_cost`	0.501	0.471	0.472	0.890	0.945	400
5	Economic benefits of climate action	`eco_benefit`	0.462	0.436	0.433	0.905	0.953	400
6	Carbon footprint of the economic sector	`eco_footprint`	0.354	0.441	0.437	0.820	0.910	400
Health Frame								
7	**Health Frame (Primary Category)**	`health_frame`	0.797	0.865	0.865	0.975	0.988	400
8	Negative impacts of climate change on health	`health_neg_impact`	0.790	0.851	0.851	0.975	0.988	400
9	Health co-benefits of climate action	`health_cobenefit`	0.304	-0.004	-0.004	0.980	0.990	400
Security Frame								
10	**Security Frame (Primary Category)**	`security_frame`	0.777	0.851	0.851	0.975	0.988	400
11	Presence of climate refugees	`security_refugees`	0.826	0.907	0.907	0.990	0.995	400

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Supplementary Table S9 – Continued from previous page

#	Category	Code	Cohen's κ	Blind phase (sentences 601–1000)				n
			(full)	κ	Krippendorff α	Gwet AC1	% agree	
12	Conflict	security_conflict	0.844	1.000	1.000	1.000	1.000	400
13	Post-disaster military assistance	security_military	0.283	-0.003	-0.001	0.990	0.995	400
14	Disruption of military operations	security_disruption	0.000	–	1.000	1.000	1.000	400
Justice Frame								
15	**Justice Frame (Primary Category)**	justice_frame	0.638	0.720	0.719	0.890	0.945	400
16	Winners and losers of climate action	justice_winners	0.275	0.386	0.381	0.925	0.963	400
17	Differentiated responsibility	justice_responsibility	0.637	0.655	0.654	0.950	0.975	400
18	Unequal vulnerability to climate change	justice_vulnerability	0.676	0.672	0.672	0.945	0.973	400
19	Unequal access to climate action	justice_access	0.058	0.168	0.160	0.910	0.955	400
20	Intergenerational justice	justice_intergen	0.645	0.731	0.731	0.975	0.988	400
Political Frame								
21	**Political Frame (Primary Category)**	political_frame	0.632	0.664	0.662	0.700	0.850	400
22	Policy action	pol_action	0.445	0.341	0.340	0.825	0.912	400
23	Political debate	pol_debate	0.498	0.542	0.534	0.615	0.807	400
24	Political positioning	pol_position	0.248	0.191	0.187	0.890	0.945	400
25	Public opinion data	pol_opinion	0.709	0.855	0.855	0.990	0.995	400
Scientific Frame								
26	**Scientific Frame (Primary Category)**	scientific_frame	0.579	0.564	0.562	0.770	0.885	400
27	Scientific debate	sci_debate	0.318	0.402	0.393	0.890	0.945	400

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Supplementary Table S9 – Continued from previous page

#	Category	Code	Cohen's κ (full)	Blind phase (sentences 601–1000) κ	Krippendorff α	Gwet AC1	% agree	n
28	Popularisation or scientific discovery	`sci_discovery`	0.506	0.518	0.517	0.805	0.902	400
29	Questioning of climate science	`sci_skepticism`	0.290	0.196	0.191	0.960	0.980	400
30	Defense of climate science	`sci_defense`	0.661	0.748	0.748	0.990	0.995	400
Environmental Frame								
31	**Environmental Frame (Primary Category)**	`environmental_frame`	0.505	0.588	0.587	0.905	0.953	400
32	Loss of natural environments	`env_habitat`	0.358	0.450	0.448	0.910	0.955	400
33	Loss of fauna and flora	`env_species`	0.410	0.432	0.430	0.915	0.958	400
Cultural Frame								
34	**Cultural Frame (Primary Category)**	`cultural_frame`	0.594	0.757	0.758	0.955	0.978	400
35	Artistic representation	`cult_art`	0.696	0.792	0.792	0.970	0.985	400
36	Difficulty to host cultural or sports events	`cult_event_impact`	0.000	0.000	0.993	0.985	0.993	400
37	Loss of indigenous practices	`cult_indigenous`	0.665	1.000	1.000	1.000	1.000	400
38	Carbon footprint of the cultural and sports sectors	`cult_footprint`	0.666	–	1.000	1.000	1.000	400
PRIMARY CATEGORIES								
Actors/Messengers								
39	**Presence of Messengers (Primary Category)**	`messenger`	0.712	0.801	0.801	0.810	0.905	400
40	Health expert	`msg_health`	0.802	0.762	0.762	0.970	0.985	400
41	Economic expert	`msg_economic`	0.680	0.652	0.653	0.905	0.953	400
42	Security expert	`msg_security`	0.490	0.460	0.460	0.955	0.978	400
43	Legal expert	`msg_legal`	0.686	0.886	0.886	0.990	0.995	400
44	Cultural or Sport expert	`msg_cultural`	0.666	0.699	0.699	0.940	0.970	400
45	Natural scientist	`msg_scientist`	0.607	0.692	0.691	0.885	0.943	400

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Supplementary Table S9 – Continued from previous page

#	Category	Code	Cohen's κ	Blind phase (sentences 601–1000)				n
			(full)	κ	Krippendorff α	Gwet AC1	% agree	
46	Social scientist	`msg_social`	0.568	0.653	0.653	0.945	0.973	400
47	Activist	`msg_activist`	0.621	0.560	0.561	0.895	0.948	400
48	Public official	`msg_official`	0.609	0.636	0.636	0.755	0.877	400
Events								
49	**Presence of Events (Primary Category)**	`event`	0.667	0.626	0.626	0.630	0.815	400
50	Extreme meteorological event	`evt_weather`	0.700	0.718	0.718	0.945	0.973	400
51	Meeting	`evt_meeting`	0.669	0.762	0.762	0.925	0.963	400
52	Publication	`evt_publication`	0.778	0.807	0.807	0.925	0.963	400
53	Election	`evt_election`	0.732	0.678	0.678	0.955	0.978	400
54	New policy	`evt_policy`	0.502	0.398	0.397	0.785	0.892	400
55	Judiciary decision	`evt_judiciary`	0.810	0.828	0.828	0.980	0.990	400
56	Cultural or Sports event	`evt_cultural`	0.737	0.632	0.631	0.945	0.973	400
57	Protest	`evt_protest`	0.749	0.805	0.805	0.970	0.985	400
Solutions								
58	**Presence of Solutions (Primary Category)**	`solution`	0.699	0.650	0.649	0.785	0.892	400
59	Mitigation strategy	`sol_mitigation`	0.712	0.661	0.661	0.820	0.910	400
60	Adaptation strategy	`sol_adaptation`	0.636	0.556	0.556	0.940	0.970	400
EMOTIONAL TONE								
61	Emotion: positive	`tone_positive`	0.454	0.468	0.468	0.830	0.915	400
62	Emotion: negative	`tone_negative`	0.543	0.561	0.561	0.590	0.795	400
63	Emotion: neutral	`tone_neutral`	0.441	0.468	0.468	0.470	0.735	400
GEOGRAPHIC FOCUS								
64	Mention of Canada	`canada`	0.876	0.870	0.870	0.870	0.935	400
URGENCY TO ACT								
65	Urgency to act	`urgency`	0.176	0.083	0.061	0.895	0.948	400

Note. Cohen's κ, Krippendorff's α, Gwet's AC1, and percent agreement are computed between the expert annotator (gold standard) and the independent second annotator. Only Cohen's κ is reported in both forms—full-sample (all 1,000 doubly-coded sentences) and blind-phase; the other indices are reported for the blind phase only (sentences 601–1000, coded without any communication with the expert), which is the primary reliability benchmark used throughout the paper. Categories with "–" in the blind-phase columns had no positive examples in sentences 601–1000.

Supplementary Table S10. Reliability tier assignment for each of the 65 annotation categories.

#	Category	Code	Validation F_1 macro			Blind	Tier			Flag
			All	**EN**	**FR**	κ	**All**	**EN**	**FR**	
MAIN FRAMES										
Economic Frame										
1	**Economic Frame (Primary Category)**	`economic_frame`	0.830	0.847	0.814	0.625	**A**	A	A	
2	Negative impacts of climate change on the economy	`eco_neg_impact`	0.783	0.820	0.741	0.542	**B**	B	B	
3	Positive impacts of climate change on the economy	`eco_pos_impact`	0.684	0.806	0.508	0.831	**B**	A	C	
4	Economic disadvantages of climate action	`eco_cost`	0.723	0.741	0.702	0.471	**B**	B	B	
5	Economic benefits of climate action	`eco_benefit`	0.848	0.885	0.817	0.436	**B**	B	B	
6	Carbon footprint of the economic sector	`eco_footprint`	0.787	0.785	0.788	0.441	**B**	B	B	
Health Frame										
7	**Health Frame (Primary Category)**	`health_frame`	0.808	0.800	0.817	0.865	**A**	A	A	
8	Negative impacts of climate change on health	`health_neg_impact`	0.364	0.386	0.340	0.851	**C**	C	C	
9	Health co-benefits of climate action	`health_cobenefit`	0.186	0.252	0.103	-0.004	**C**	C	C	
Security Frame										
10	**Security Frame (Primary Category)**	`security_frame`	0.782	0.825	0.730	0.851	**B**	A	B	
11	Presence of climate refugees	`security_refugees`	0.481	0.476	0.417	0.907	**C**	C	C	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind	Tier			Flag
			All	EN	FR	κ	All	EN	FR	
12	Conflict	`security_conflict`	0.324	0.358	0.286	1.000	**C**	C	C	
13	Post-disaster military assistance	`security_military`	0.045	0.000	0.045	-0.003	**C**	C	C	excluded
14	Disruption of military operations	`security_disruption`	1.000	0.000	1.000	–	**C**	C	C	excluded
Justice Frame										
15	**Justice Frame (Primary Category)**	`justice_frame`	0.857	0.832	0.880	0.720	**A**	A	A	
16	Winners and losers of climate action	`justice_winners`	0.708	0.654	0.737	0.386	**C**	C	C	PIRD*
17	Differentiated responsibility	`justice_responsibility`	0.860	0.900	0.823	0.655	**A**	A	A	
18	Unequal vulnerability to climate change	`justice_vulnerability`	0.845	0.786	0.894	0.672	**A**	B	A	
19	Unequal access to climate action	`justice_access`	0.651	0.475	0.796	0.168	**C**	C	C	
20	Intergenerational justice	`justice_intergen`	0.942	0.924	0.952	0.731	**A**	A	A	
Political Frame										
21	**Political Frame (Primary Category)**	`political_frame`	0.829	0.812	0.842	0.664	**A**	A	A	
22	Policy action	`pol_action`	0.756	0.802	0.722	0.341	**C**	C	C	PIRD*
23	Political debate	`pol_debate`	0.564	0.469	0.650	0.542	**C**	C	B	
24	Political positioning	`pol_position`	0.782	0.752	0.808	0.191	**C**	C	C	PIRD*
25	Public opinion data	`pol_opinion`	0.894	0.877	0.910	0.855	**A**	A	A	
Scientific Frame										
26	**Scientific Frame (Primary Category)**	`scientific_frame`	0.890	0.891	0.890	0.564	**B**	B	B	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind	Tier			Flag
			All	**EN**	**FR**	κ	**All**	**EN**	**FR**	
27	Scientific debate	`sci_debate`	0.911	0.962	0.867	0.402	**B**	B	B	
28	Popularisation or scientific discovery	`sci_discovery`	0.838	0.884	0.797	0.518	**B**	B	B	
29	Questioning of climate science	`sci_skepticism`	0.775	0.783	0.768	0.196	**C**	C	C	PIRD*
30	Defense of climate science	`sci_defense`	0.682	0.769	0.460	0.748	**B**	B	C	
Environmental Frame										
31	**Environmental Frame (Primary Category)**	`environmental_frame`	0.871	0.866	0.875	0.588	**B**	B	B	
32	Loss of natural environments	`env_habitat`	0.721	0.804	0.573	0.450	**B**	B	C	
33	Loss of fauna and flora	`env_species`	0.703	0.714	0.692	0.432	**B**	B	B	
Cultural Frame										
34	**Cultural Frame (Primary Category)**	`cultural_frame`	0.708	0.802	0.581	0.757	**B**	A	C	
35	Artistic representation	`cult_art`	0.659	0.834	0.469	0.792	**B**	A	C	
36	Difficulty to host cultural or sports events	`cult_event_impact`	0.484	0.631	0.354	0.000	**C**	C	C	
37	Loss of indigenous practices	`cult_indigenous`	0.595	0.824	0.415	1.000	**C**	A	C	
38	Carbon footprint of the cultural and sports sectors	`cult_footprint`	0.351	0.702	0.014	–	**C**	B	C	
PRIMARY CATEGORIES										
Actors/Messengers										
39	**Presence of Messengers (Primary Category)**	`messenger`	0.961	0.955	0.967	0.801	**A**	A	A	
40	Health expert	`msg_health`	0.907	0.862	0.951	0.762	**A**	A	A	
41	Economic expert	`msg_economic`	0.874	0.847	0.895	0.652	**A**	A	A	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind	Tier			Flag
			All	**EN**	**FR**	κ	**All**	**EN**	**FR**	
42	Security expert	`msg_security`	0.787	0.728	0.813	0.460	**B**	B	B	
43	Legal expert	`msg_legal`	0.914	0.861	0.927	0.886	**A**	A	A	
44	Cultural or Sport expert	`msg_cultural`	0.843	0.853	0.825	0.699	**A**	A	A	
45	Natural scientist	`msg_scientist`	0.907	0.920	0.896	0.692	**A**	A	A	
46	Social scientist	`msg_social`	0.944	0.938	0.948	0.653	**A**	A	A	
47	Activist	`msg_activist`	0.918	0.934	0.903	0.560	**B**	B	B	
48	Public official	`msg_official`	0.929	0.927	0.930	0.636	**A**	A	A	
Events										
49	**Presence of Events (Primary Category)**	`event`	0.897	0.881	0.912	0.626	**A**	A	A	
50	Extreme meteorological event	`evt_weather`	0.889	0.887	0.891	0.718	**A**	A	A	
51	Meeting	`evt_meeting`	0.877	0.855	0.891	0.762	**A**	A	A	
52	Publication	`evt_publication`	0.958	0.950	0.962	0.807	**A**	A	A	
53	Election	`evt_election`	0.903	0.911	0.895	0.678	**A**	A	A	
54	New policy	`evt_policy`	0.829	0.821	0.837	0.398	**C**	C	C	PIRD*
55	Judiciary decision	`evt_judiciary`	0.808	0.628	0.939	0.828	**A**	B	A	
56	Cultural or Sports event	`evt_cultural`	0.881	0.906	0.857	0.632	**A**	A	A	
57	Protest	`evt_protest`	0.943	0.982	0.909	0.805	**A**	A	A	
Solutions										
58	**Presence of Solutions (Primary Category)**	`solution`	0.900	0.891	0.909	0.650	**A**	A	A	
59	Mitigation strategy	`sol_mitigation`	0.839	0.872	0.801	0.661	**A**	A	A	
60	Adaptation strategy	`sol_adaptation`	0.895	0.889	0.903	0.556	**B**	B	B	
EMOTIONAL TONE										
61	Emotion: positive	`tone_positive`	0.702	0.712	0.692	0.468	**B**	B	B	
62	Emotion: negative	`tone_negative`	0.799	0.793	0.804	0.561	**B**	B	B	
63	Emotion: neutral	`tone_neutral`	0.777	0.756	0.798	0.468	**B**	B	B	

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Supplementary Table S10 – Continued from previous page

#	Category	Code	Validation F_1 macro			Blind	Tier			Flag
			All	**EN**	**FR**	κ	**All**	**EN**	**FR**	
GEOGRAPHIC FOCUS										
64	Mention of Canada	`canada`	0.982	0.986	0.978	0.870	**A**	A	A	
URGENCY TO ACT										
65	Urgency to act	`urgency`	0.853	0.874	0.835	0.083	**C**	C	C	PIRD*
TIER COUNTS: All: A=27 / B=21 / C=17; EN: A=30 / B=20 / C=15; FR: A=27 / B=17 / C=21; PIRD*: 6; excluded from training: 2.										

Note. Each category is assigned to a reliability tier based on its validation F_1 macro and the blind-phase Cohen's κ. **Tier A** (research-grade) requires $F_1 \geq 0.80$ *and* $\kappa \geq 0.60$; **Tier C** (exploratory) corresponds to $F_1 < 0.60$ *or* $\kappa < 0.40$; **Tier B** (use with caution) captures everything in between. The acronym **PIRD*** in the Flag column stands for *prevalence-induced reliability deflation*: it identifies categories with $F_1 \geq 0.70$ but blind-phase Cohen's $\kappa < 0.40$, where the chance-corrected agreement is depressed by very low prevalence rather than by systematic disagreement between coders [18, 19]. The two categories excluded from training due to insufficient positive examples are flagged "excluded" and forced to Tier C in every language column.

Supplementary Table S11. Per-model training provenance: best epoch, training phase, validation macro F_1, and positive-class support for every BERT/CamemBERT classifier.

#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	**Training phase**	F_1	n^+	**Epoch**	**Training phase**	F_1	n^+
MAIN FRAMES										
Economic Frame										
1	**Economic Frame (Primary Category)**	`economic_frame`	4	r	0.845	24	13	n	0.885	28
2	Negative impacts of climate change on the economy	`eco_neg_impact`	12	n	0.821	6	3	n	0.917	8
3	Positive impacts of climate change on the economy	`eco_pos_impact`	6	n	0.664	5	4	r	0.698	3
4	Economic disadvantages of climate action	`eco_cost`	5	n	0.732	6	1	r	0.707	4
5	Economic benefits of climate action	`eco_benefit`	9	n	0.726	3	17	n	0.749	16
6	Carbon footprint of the economic sector	`eco_footprint`	8	n	0.897	7	16	n	0.904	8
Health Frame										
7	**Health Frame (Primary Category)**	`health_frame`	9	n	0.894	6	6	n	0.830	4
8	Negative impacts of climate change on health	`health_neg_impact`	1	n	0.455	5	1	n	0.429	3
9	Health co-benefits of climate action	`health_cobenefit`	17	r	0.486	1	2	r	0.119	2
Security Frame										
10	**Security Frame (Primary Category)**	`security_frame`	9	n	0.933	12	11	n	0.898	2
11	Presence of climate refugees	`security_refugees`	2	n	1.000	1	9	r	1.000	1
12	Conflict	`security_conflict`	1	n	1.000	1	14	r	1.000	1
13	Post-disaster military assistance	`security_military`	–	–	–	–	3	r	0.026	2
14	Disruption of military operations	`security_disruption`	–	–	–	–	18	n	1.000	1
Justice Frame										
15	**Justice Frame (Primary Category)**	`justice_frame`	5	n	0.847	32	10	n	0.849	30

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Supplementary Table S11 – Continued from previous page

#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	**Training phase**	F_1	n^+	**Epoch**	**Training phase**	F_1	n^+
16	Winners and losers of climate action	`justice_winners`	13	n	0.800	2	14	n	0.795	2
17	Differentiated responsibility	`justice_responsibility`	4	n	0.883	4	3	n	0.750	4
18	Unequal vulnerability to climate change	`justice_vulnerability`	19	r	0.782	6	13	n	0.798	5
19	Unequal access to climate action	`justice_access`	13	r	0.182	2	4	r	0.809	9
20	Intergenerational justice	`justice_intergen`	9	n	0.966	7	11	n	0.899	3
Political Frame										
21	**Political Frame (Primary Category)**	`political_frame`	3	n	0.853	46	5	n	0.831	133
22	Policy action	`pol_action`	2	n	0.803	17	4	n	0.816	6
23	Political debate	`pol_debate`	7	n	0.569	38	4	n	0.768	43
24	Political positioning	`pol_position`	5	n	0.848	6	4	n	0.895	3
25	Public opinion data	`pol_opinion`	4	n	0.954	6	12	n	1.000	2
Scientific Frame										
26	**Scientific Frame (Primary Category)**	`scientific_frame`	3	n	0.869	27	5	n	0.832	56
27	Scientific debate	`sci_debate`	5	n	0.793	10	7	n	0.688	5
28	Popularisation or scientific discovery	`sci_discovery`	3	n	0.814	14	6	n	0.844	16
29	Questioning of climate science	`sci_skepticism`	3	r	0.827	5	5	r	0.664	4
30	Defense of climate science	`sci_defense`	6	n	0.752	4	1	n	0.459	3
Environmental Frame										
31	**Environmental Frame (Primary Category)**	`environmental_frame`	7	n	0.915	9	13	r	0.802	6
32	Loss of natural environments	`env_habitat`	7	n	1.000	5	5	n	0.829	3
33	Loss of fauna and flora	`env_species`	5	n	0.873	4	3	n	0.829	3
Cultural Frame										
34	**Cultural Frame (Primary Category)**	`cultural_frame`	7	n	0.879	24	13	r	0.913	5
35	Artistic representation	`cult_art`	2	n	1.000	2	7	n	1.000	2
36	Difficulty to host cultural or sports events	`cult_event_impact`	18	n	0.850	7	4	n	1.000	2

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Supplementary Table S11 – Continued from previous page

#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	Training phase	F_1	n^+	Epoch	Training phase	F_1	n^+
37	Loss of indigenous practices	`cult_indigenous`	6	n	1.000	1	3	r	0.462	2
38	Carbon footprint of the cultural and sports sectors	`cult_footprint`	7	n	1.000	1	3	r	0.074	1
PRIMARY CATEGORIES										
Actors/Messengers										
39	**Presence of Messengers (Primary Category)**	`messenger`	11	n	0.908	72	6	n	0.922	227
40	Health expert	`msg_health`	9	r	0.927	6	8	n	0.954	6
41	Economic expert	`msg_economic`	4	n	0.785	6	3	r	0.861	7
42	Security expert	`msg_security`	8	n	0.830	2	9	n	1.000	1
43	Legal expert	`msg_legal`	4	n	0.833	2	11	n	0.883	6
44	Cultural or Sport expert	`msg_cultural`	3	n	1.000	1	3	r	0.773	9
45	Natural scientist	`msg_scientist`	8	n	0.883	37	8	n	0.902	12
46	Social scientist	`msg_social`	7	n	0.883	7	16	r	0.816	12
47	Activist	`msg_activist`	5	n	0.775	5	5	n	1.000	6
48	Public official	`msg_official`	4	n	0.800	18	10	n	0.943	24
Events										
49	**Presence of Events (Primary Category)**	`event`	11	n	0.865	33	10	n	0.876	113
50	Extreme meteorological event	`evt_weather`	3	n	1.000	8	7	n	0.912	6
51	Meeting	`evt_meeting`	8	n	0.880	8	3	n	0.969	11
52	Publication	`evt_publication`	4	n	0.931	25	7	n	1.000	12
53	Election	`evt_election`	4	n	1.000	1	10	r	0.899	5
54	New policy	`evt_policy`	7	n	0.856	6	8	n	0.841	6
55	Judiciary decision	`evt_judiciary`	7	n	1.000	1	9	n	1.000	1
56	Cultural or Sports event	`evt_cultural`	4	n	0.664	5	7	n	1.000	1
57	Protest	`evt_protest`	4	n	0.944	5	5	n	0.862	6
Solutions										
58	**Presence of Solutions (Primary Category)**	`solution`	6	n	0.825	90	4	n	0.911	46
59	Mitigation strategy	`sol_mitigation`	16	n	0.842	79	7	n	0.877	105
60	Adaptation strategy	`sol_adaptation`	9	n	0.843	15	10	n	0.894	21
EMOTIONAL TONE										
61	Emotion: positive	`tone_positive`	8	n	0.746	11	6	n	0.829	16
62	Emotion: negative	`tone_negative`	2	n	0.745	152	16	n	0.804	158

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Supplementary Table S11 – Continued from previous page

#	Category	Code	English (BERT)				French (CamemBERT)			
			Epoch	Training phase	F_1	n^+	Epoch	Training phase	F_1	n^+
63	Emotion: neutral	`tone_neutral`	4	n	0.709	205	4	n	0.741	80
GEOGRAPHIC FOCUS										
64	Mention of Canada	`canada`	7	n	0.955	136	12	n	0.972	54
URGENCY TO ACT										
65	Urgency to act	`urgency`	10	r	0.783	20	7	n	0.816	19

Summary: 128 trained models. Reinforcement triggered for 45 models; best epoch retained from the reinforced phase for 23 models. Static training configuration (optimiser, learning rate, batch size, random seed, hardware) is reported in the Methods section.

Note. The English models are BERT-base-cased (case-sensitive: load their tokeniser with `do_lower_case=False`) and the French models are CamemBERT-base, both trained through the AugmentedSocialScientist framework. The four per-language sub-columns are: **Ep.** (best epoch retained by the combined-criterion best-model selector $0.7 \cdot F_1^{(1)} + 0.3\cdot$ macro-F_1), **Ph.** (training phase — n for normal, r for reinforced training, the latter firing when the positive-class F_1 falls below 0.60 during normal training), F_1 (macro F_1 at the retained epoch), and n^+ (number of positive validation examples; "–" denotes models that failed to train because of insufficient positive examples). The static configuration (optimiser AdamW with $\epsilon = 10^{-8}$, learning rate 5×10^{-5} in normal training and 1×10^{-5} in the reinforced phase, batch size 32 normal / 64 reinforced, maximum sequence length 512 tokens, up to 20 epochs in each phase, random seed 42 for Python/NumPy/PyTorch RNGs, linear schedule with zero warm-up, weighted cross-entropy in normal training and a `WeightedRandomSampler` in the reinforced phase, training on an Apple Mac Studio M2 Ultra with 128 GB unified memory) is documented in full in the Methods section of the main manuscript.

Supplementary Table S12. Complete data dictionary of the enriched CCF Database: every column of every table, its type, and a one-line definition.

Column	Type	Definition
Table CCF_full_data *(article-level metadata, 283,964 rows)*		
`doc_id`	BIGINT (PK)	Unique article identifier.
`news_type`	TEXT	Editorial category (news, opinion, editorial, etc.).
`title`	TEXT	Article headline.
`author`	TEXT	Author byline when available.
`media`	TEXT	Newspaper name.
`words_count`	BIGINT	Total word count of the article.
`date`	DATE	Publication date.
`language`	TEXT	"EN" or "FR".
`page_number`	TEXT	Print page (free text, e.g. "A1" or "B3").
Table CCF_processed_data *(sentence-level annotations, 9,908,776 rows)*		
`doc_id`	BIGINT	Foreign key to CCF_full_data.
`sentence_id`	BIGINT	Sentence position within the article (starting at 1; 0 reserved for title in the embeddings table).
Ten article-level metadata columns are duplicated here for query convenience (news_type, title, author, media, words_count, date, language, page_number).		
65 annotation columns	INTEGER (0/1, nullable)	One binary flag per category from Supplementary Table S3. Sub-categories are NULL when their parent primary detection is negative.
`ner_entities`	TEXT (JSON)	Per-sentence NER output as `{"PER": [..], "ORG": [..], "LOC": [..]}`.
Table CCF_article_aggregates *(article-level rollup, 283,964 rows)*		
`doc_id`	BIGINT (PK)	Foreign key to CCF_full_data.
`n_sentences`	INTEGER	Total number of two-sentence analytical units in the article.
`n_sentences_en`	INTEGER	Subset in English.
`n_sentences_fr`	INTEGER	Subset in French.
65 `prop_X` columns	DOUBLE PRECISION ($[0, 1]$)	Article-level proportion of sentences positive for category X (sum / n_sentences, with NULL sub-categories coalesced to 0).
`top_frame`	TEXT	Main frame with the highest `prop_X`, or NULL for articles with no positive frame.
`top_frame_prop`	DOUBLE PRECISION	Proportion of the top frame.
`entropy_frames`	DOUBLE PRECISION	Shannon entropy (natural base) of the eight main-frame proportions; 0 when the article has no positive frame.

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Column	Type	Definition
`n_unique_messenger_subs`	SMALLINT	Number of distinct messenger sub-categories present in the article (0–9).
`n_unique_event_subs`	SMALLINT	Same for event sub-categories (0–8).
`n_unique_solution_subs`	SMALLINT	Same for solution sub-categories (0–2).
Table CCF_article_entities *(NER rollup, 283,964 rows)*		
`doc_id`	BIGINT (PK)	Foreign key to CCF_full_data.
`entities_per`	JSONB	Deduplicated array of persons mentioned in the article.
`entities_org`	JSONB	Deduplicated array of organisations.
`entities_loc`	JSONB	Deduplicated array of locations.
`n_unique_per`	INTEGER	Cardinality of `entities_per`.
`n_unique_org`	INTEGER	Cardinality of `entities_org`.
`n_unique_loc`	INTEGER	Cardinality of `entities_loc`.
`n_sentences_with_per`	INTEGER	Number of sentences with at least one PER entity.
`n_sentences_with_org`	INTEGER	Same for ORG.
`n_sentences_with_loc`	INTEGER	Same for LOC.
Table CCF_reliability_tiers *(per-category quality lookup, 65 rows)*		
`number`	SMALLINT (PK)	Table B1 number (1–65).
`category`	TEXT	Readable category name.
`code`	TEXT (UNIQUE)	Column code as used in CCF_processed_data.
`tier_overall`	CHAR(1)	Tier on the combined corpus: "A", "B", or "C".
`tier_en`	CHAR(1)	Tier on the English subset.
`tier_fr`	CHAR(1)	Tier on the French subset.
`paradox_kappa`	BOOLEAN	True if the category exhibits prevalence-induced reliability deflation, i.e. $F_1 \geq 0.70$ and $\kappa < 0.40$. The column name is kept for backwards compatibility with the original schema.
`exclusion_reason`	TEXT	"insufficient_training_data" for the two categories that could not be trained, NULL otherwise.

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Supplementary Table S12 – Continued from previous page

Column	**Type**	**Definition**
`f1_macro_all`, `f1_macro_en`, `f1_macro_fr`	DOUBLE PRECISION	Validation macro F_1 score on the gold standard.
`support_all`, `support_en`, `support_fr`	INTEGER	Number of positive (class 1) validation examples.
`cohens_kappa_blind_phase`, `gwet_ac1_blind_phase`, `krippendorff_alpha_blind_phase`	DOUBLE PRECISION	Inter-coder agreement on the blind phase (sentences 601–1000).
`n_blind_phase`	INTEGER	Sentence count of the blind phase.
`prev_coder1_full`, `prev_coder2_full`	DOUBLE PRECISION	Prevalence (proportion of positives) per coder on the full 1,000-sentence sample.
Table CCF_sentence_embeddings *(BGE-M3 dense vectors, 10,192,740 rows)*		
`doc_id`	BIGINT (PK, with `sentence_id`)	Foreign key to CCF_full_data.
`sentence_id`	INTEGER (PK)	Sentence position within the article. `sentence_id` = 0 is reserved for the article title.
`embedding`	`halfvec(1024)`	L2-normalised BGE-M3 vector. `pgvector` cosine distance via `<=>`; HNSW index on cosine ops.

Note. All annotation column codes (`economic_frame`, `eco_neg_impact`, etc.) and their definitions appear in Supplementary Table S3.

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